

Quantum Computing & Hilbert Space

Slide 1: State Vectors & Unitary Evolution

Aurora Notes - Presentation Deck

Qubits & Superposition

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

Probabilistic measurement amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$

Summary & Next Steps

1. Variational Quantum Circuits
2. Parameter Shift Rule